

Ex. 25. Simplify : $99\frac{48}{49} \times 245$.

Sol. Given expression = $\left(100 - \frac{1}{49}\right) \times 245 = \frac{4899}{49} \times 245 = 4899 \times 5 = 24495$.

Ex. 26. Find the value of: $\frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110}$.

(A.A.O. Exam, 2009)

Sol. Given expression = $\left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \left(\frac{1}{7} - \frac{1}{8}\right) + \left(\frac{1}{8} - \frac{1}{9}\right) + \left(\frac{1}{9} - \frac{1}{10}\right) + \left(\frac{1}{10} - \frac{1}{11}\right)$
 $= \left(\frac{1}{5} - \frac{1}{11}\right) = \frac{6}{55}$.

Ex. 27. A board 7 ft. 9 inches long is divided into 3 equal parts. What is the length of each part?

Sol. Length of board = 7 ft. 9 inches = $(7 \times 12 + 9)$ inches = 93 inches.

$\therefore$ Length of each part = $\left(\frac{93}{3}\right)$ inches = 31 inches = 2 ft. 7 inches.

Ex. 28. Ram went to a shop to buy 50 kg of rice. He buys two varieties of rice which cost him ₹ 4.50 per kg and ₹ 5 per kg. He spends a total of ₹ 240. What was the quantity of rice bought which cost him ₹ 4.50 per kg?

(M.A.T., 2007)

Sol. Let the quantity of rice bought at ₹ 4.50 per kg be x kg.

Then, quantity of rice bought at ₹ 5 per kg = $(50 - x)$ kg.

$\therefore 4.50x + 5(50 - x) = 240 \Rightarrow 250 - 0.5x = 240 \Rightarrow 0.5x = 10 \Rightarrow x = 20$.

Hence, quantity of rice bought at ₹ 4.50 per kg = 20 kg.

Ex. 29. A boy was asked to multiply a certain number by 53. He multiplied it by 35 and got his answer less than the correct one by 1206. Find the number to be multiplied.

(SNAP, 2005)

Sol. Let the required number be x .

Then, $53x - 35x = 1206 \Leftrightarrow 18x = 1206 \Leftrightarrow x = \frac{1206}{18} = 67$.

Hence, number to be multiplied = 67.

Ex. 30. $\frac{4}{15}$ of $\frac{5}{7}$ of a number is greater than $\frac{4}{9}$ of $\frac{2}{5}$ of the same number by 8. What is half of that number?

Sol. Let the number be x .

Then, $\frac{4}{15}$ of $\frac{5}{7}$ of $x - \frac{4}{9}$ of $\frac{2}{5}$ of $x = 8 \Leftrightarrow \frac{4}{21}x - \frac{8}{45}x = 8$

$\Leftrightarrow \left(\frac{4}{21} - \frac{8}{45}\right)x = 8 \Leftrightarrow \left(\frac{60 - 56}{315}\right)x = 8 \Leftrightarrow \frac{4}{315}x = 8$

$\Leftrightarrow x = \left(\frac{8 \times 315}{4}\right) = 630 \Leftrightarrow \frac{1}{2}x = 315$.

Hence, required number = 315.

Ex. 31. A man owns $\frac{2}{3}$ of the market research bureau business and sells $\frac{3}{4}$ of his shares for ₹ 75000. What is the value of business?

(Campus Recruitment, 2010)

Sol. Let the total value be ₹ x .

Then, $\frac{3}{4}$ of $\frac{2}{3}$ of $x = 75000 \Leftrightarrow \frac{x}{2} = 75000 \Leftrightarrow x = 150000$.

$\therefore$ Value of business = ₹ 150000.